

NUTRIENT	ACTION IN COVID PREVENTION AND TREATMENT
Vitamin E	Supports integrity of epithelial membranes, increases IL-2, NK cell activity, T cells proliferation & antibody response. Vitamin E & Selenium exhibit strong control over viral replication and mutation.
Zinc	Improves immune response, nucleic acid synthesis & redox homeostasis. Intracellular zinc impairs virus replication. Zinc deficiency enhances production of proinflammatory cytokines.
Selenium	Selenium deficiency & low antioxidants encourages pathogenic mutation of the virus. Selenium deficiency impairs functions of neutrophils, T cells, lymphocytes, NK cells & thymocytes.
Iodine	Iodide secreted in the surface liquid of the upper airway has antiviral effect. Direct exposure to iodine can destroy the virus.
Copper	Disintegrates the viral envelope & destroys the virus, also has anti viral effect by binding electron donor groups on viral proteins & nucleic acids.
Iron	Iron deficiency anaemia reduces immune response and increases susceptibility to viral infection. Iron reduces NF-kB nuclear translocation, promotes M2 & decreases proinflammatory M1 macrophage activity. Iron regulates cytokines via hepcidin, iron oxides disintegrate the virus lipid envelope by peroxidation.
Vitamin A	Antioxidant, main regulator of mucosal immunity, inhibits viral replication, promotes immune response, decreases morbidity & mortality.
Vitamin B-complex	Vitamins B1, B2, and B5 control host immune response through regulation of energy generation in immune cells. Down regulate proinflammatory cytokines & NF-kB, & activate macrophages.
Vitamin C	Vitamin C inhibits proinflammatory cytokines, like TNF and IL-6, promotes production of interferon, proliferation and migration of immune cells, enhances antibodies and complement proteins. Vitamin C is viricidal, in high doses.

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